Experiment-11

Develope customized Ping command to test server connectivity.

Aim:

To develope a customized Ping command using Python to test the connectivity and response time.

Procedure:

1) import socket, struct, os, time and select modules.
2) create a function to calculate icmp checksum.
3) Build ICMP echo request
4) create raw socket with socket ipPort
5) send the packet to the target hosting using sendto()
6) for persistent configuration in Linux /etc/network/interfaces or user mode

Command / code

Sudo ip addr add 192.168.1.1/24

Sudo ip route add default via 192.168.1.1

ip a show eth0

ping 192.168.1.1

nmcli con mod "wired connection 1" ipv4.

nmcli con mod "wired connection 1" gateway

nmcli con mod "wired connection 1" ipv4.

nmcli con up "wired connection 1"

7) Verify the configuration using 'ipconfig' or 'ip a'.

8) Ping the default gateway to confirm network connectivity.

9) Ping another computer on the same subnet to verify communication.

Result:

Static IP address were successfully assigned to computer. Connected was verified using ping commands between hosts on the same subnet.